

DroidLink WebUI Sequence Builder How-To

(DroidLink_Slave + DL_AP)

This guide is written for builders who are new to sequence logic.

1) What a Sequence Builder actually is

A sequence builder is a timeline editor.

It does not automatically "figure out" a routine for you.

It only runs the exact actions you put on the timeline.

Think of each action like this:

- At time X, tell target Y to do action Z.

If you do not add an action, that behavior will not happen.

2) What an action actually does

An action is a single command.

Example:

- "Open Drawer at 0 ms" = one command.

That command does not include "close later." If you want the drawer to close, you must add a second action with a later start time.

Example timeline:

- Action 1: Start 0 ms -> Drawer -> Open
- Action 2: Start 3000 ms -> Drawer -> Close

This means:

- Open now
 - Wait about 3 seconds
 - Then close
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3) Beginner mental model (always use this)

Every action answers 4 questions:

1. When does it start? (Start ms)
2. What does it control? (Target output/group/LED section)
3. What should it do? (Open/Close/Position/Animation/Off/etc.)
4. How long or until what condition? (Move time, timeout, stop input)

If one of those is missing, behavior is usually confusing or unsafe.

4) Why beginners think sequences are broken

"I opened it, but it didn't close"

Because closing is a different command. Add another action later.

"My 360 servo keeps spinning"

Forward/Reverse keeps running until a Stop action (or other stopping condition). Add a Stop action at a later time.

"I turned an output on and it stayed on"

On and Off are separate actions unless you use an input-controlled action that auto-returns off.

5) Build a lights-only sequence (LED Seq)

1. Open LED Seq.
2. Click Add Action.
3. For each action set:
 - Start time (ms)
 - LED section (or Entire strip)
 - Action type (animation / solid color / off)
 - Effect options (color, speed, brightness)
4. Repeat until all lighting actions are added.
5. Click Preview sequence.
6. Enter Sequence name.
7. (Optional) assign a command shortcut.
8. Click Save sequence and command.

Important: each LED action is also a timed command. Lighting changes only when the timeline reaches that action.

6) Build a servo sequence (Seq Builder)

1. Open Seq Builder.
2. Click Add sequence action.
3. For each action set:
 - Start ms
 - Target output or group
 - Action type (Open, Close, Position, Flutter, etc.)
 - Timing/safety options when shown
4. Add all movement actions needed.
5. Click Run sequence to test.
6. Use Stop any time for safety.

7. Enter Sequence name and command slot.
8. Click Save Sequence.

Important: a servo sequence is not a script language. It is a timed list of direct commands.

7) Build a combined servo + lights sequence

1. First, build and save an LED sequence in LED Seq.
2. Build your movement actions in Seq Builder.
3. In Add a saved LED sequence (optional), select the LED preset, set Start offset (0 ms = start with first movement action), and click Add LED sequence.
4. Test the combined sequence.
5. Save it as a normal sequence.

Important: attaching a saved LED sequence adds that full lighting timeline as one reusable block.

8) Command ranges (user-facing)

- 00-59: servo-only or combined servo+LED sequences
- 60: restore startup lighting
- 61: turn all configured LEDs off
- 62-91: LED-only saved sequences

Role prefix depends on configuration (BS / DS / LS / US).

9) Safety guidance to include with every builder

- Calibrate with linkages disconnected first.
 - Test one mechanism at a time.
 - Use input limits and safety timeouts where applicable.
 - Keep a hand near power during first live tests.
 - Treat every action as a direct hardware command.
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One sentence to teach every new user

A sequence is a timed list of individual commands; one action does one thing, so return motions (like close or stop) must be added as separate later actions.